3 **(a)** State the principle of superposition.

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..... [2]

(b) A source of microwaves is placed on a table at a fixed distance from a detector. A vertical reflecting plate is placed a distance y from the source and the detector. Reflection at the reflector causes a phase change of π rad to the microwave.

Fig 3.1 shows the **view from above**.

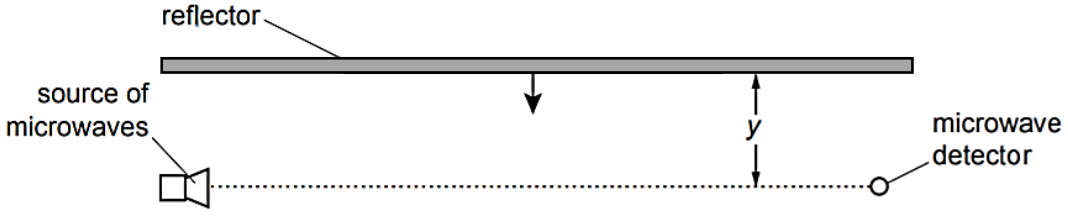

Fig 3.1 (top view)

The reflector is moved gradually towards the source and the detector, in the direction indicated by the arrow on the reflector in Fig 3.1. The intensity measured by the detector alternates between high and low.

Explain these observations.

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..... [3]

(c) The reflector is fixed at a position y = 2.5 m and the detector is at a distance 5.0 m away. The wavelength of the microwave is 0.59 m.

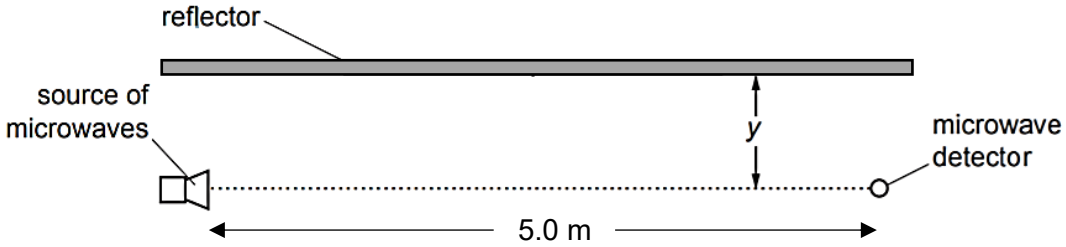

Fig. 3.2

(i) Explain whether the intensity at the detector is a maximum or a minimum.

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..... [3]

(ii) Describe how the intensity of reception varies as the detector is moved towards the source along the dotted line until it reaches the source.

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..... [3]